# HOTEL BOOKING DEMAND, CUSTOMER BEHAVIOUR AND CANCELLATION ANALYSIS

**A Data Analytics Project**

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**Dataset:** Hotel Booking Demand Dataset

**Tools and Technologies:** Python, Pandas, NumPy, Matplotlib, Seaborn and Jupyter Notebook

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## 1. Abstract

The hospitality industry relies on effective booking management, customer behaviour analysis and revenue planning to maintain operational efficiency and profitability. Understanding booking patterns, cancellation behaviour, seasonal demand and room pricing can help hotels make informed business decisions.

This project presents an exploratory data analysis (EDA) of the Hotel Booking Demand dataset, containing 119,390 booking records and 32 original variables. The analysis covers two hotel categories: City Hotel and Resort Hotel. Python libraries, including Pandas, NumPy, Matplotlib and Seaborn, were used to inspect, process, analyse and visualise the data.

The project investigates booking characteristics, customer preferences, hotel-wise performance, cancellation patterns, arrival seasonality and Average Daily Rate (ADR). It also examines relationships between cancellations and factors such as lead time, deposit type, customer type, market segment, special requests and previous cancellation history.

The analysis identifies differences between City Hotel and Resort Hotel in booking volume, cancellation rates, stay duration and room pricing. It also highlights seasonal variations in booking demand and ADR. These findings provide insights that can support hotel management decisions related to reservation policies, pricing strategies and demand planning.

## 2. Introduction

The hospitality industry generates substantial booking data through direct reservations, online travel agencies, travel agents and other distribution channels. Analysing this information can help hotels understand customer preferences, identify cancellation risks and improve revenue management.

Hotel bookings are influenced by several factors, including the type of hotel, customer category, market segment, booking lead time, deposit policy, arrival period and room pricing. Cancellation behaviour is particularly important because cancelled reservations can affect occupancy planning and expected revenue.

This project uses the Hotel Booking Demand dataset to perform a comprehensive exploratory analysis of booking records from City Hotel and Resort Hotel. The analysis focuses on understanding the characteristics of bookings, comparing the performance of the two hotel categories and identifying patterns that may be useful for hospitality management.

## 3. Objectives of the Project

The major objectives are:

1. To inspect and understand the structure and characteristics of the hotel booking dataset.
2. To identify missing values, duplicate records and unusual observations.
3. To analyse booking characteristics and customer behaviour.
4. To compare City Hotel and Resort Hotel across important booking indicators.
5. To examine cancellation patterns and identify factors associated with cancellations.
6. To investigate booking demand and seasonal trends across arrival periods.
7. To analyse ADR across hotel types, customer categories, room types and booking characteristics.
8. To identify patterns that can support better reservation management and pricing decisions.
9. To present the findings through statistical summaries and visualisations.

**4. Dataset Description**

The project uses the Hotel Booking Demand dataset, which contains booking information for two hotel categories:

- City Hotel
- Resort Hotel

The original dataset consists of **119,390 observations and 32 variables**. Each row represents a booking record rather than necessarily a unique individual customer.

The dataset contains information about booking status, arrival dates, length of stay, guest composition, meal preferences, market segments, distribution channels, room types, deposit arrangements, customer history and ADR.

**4.1 Important Variables**

| Variable | Description |
|---|---|
| hotel | Type of hotel: City Hotel or Resort Hotel |
| is_canceled | Indicates whether the booking was cancelled |
| lead_time | Number of days between booking and arrival |
| arrival_date_year | Year of arrival |
| arrival_date_month | Month of arrival |
| stays_in_weekend_nights | Weekend nights booked |
| stays_in_week_nights | Weekday nights booked |
| adults | Number of adults |
| children | Number of children |
| babies | Number of babies |
| meal | Type of meal plan selected |
| country | Country of origin of the guest |
| market_segment | Market segment associated with the booking |
| distribution_channel | Channel through which the booking was made |
| is_repeated_guest | Indicates whether the guest has booked previously |
| previous_cancellations | Number of previous bookings cancelled by the guest |
| reserved_room_type | Initially reserved room category |
| assigned_room_type | Room category assigned to the guest |
| booking_changes | Number of changes made to the booking |
| deposit_type | Deposit arrangement |
| customer_type | Type of customer |
| adr | Average Daily Rate |
| total_of_special_requests | Number of special requests made |
| reservation_status | Final reservation status |
| reservation_status_date | Date on which the reservation status was recorded |

**5. Tools and Technologies Used**

The project was conducted using Python in Jupyter Notebook.

- **Python:** Used as the primary programming language for data analysis.
- **Pandas:** Used for data loading, cleaning, transformation, grouping, aggregation and statistical analysis.
- **NumPy:** Used for numerical operations and creation of derived variables.
- **Matplotlib:** Used to create charts and visualise analytical results.
- **Seaborn:** Used for statistical visualisations, including histograms, boxplots, countplots, barplots, scatterplots and heatmaps.
- **Jupyter Notebook:** Used as the development environment to organise code, outputs, visualisations and interpretations.

**6. Methodology:** The project followed a structured exploratory data analysis workflow.

**6.1 Data Loading and Inspection**

The dataset was loaded into a Pandas DataFrame. Initial inspection included displaying sample records, checking dataset dimensions, examining column names, identifying data types and reviewing non-null values.

The original dataset contained 119,390 rows and 32 columns. Both City Hotel and Resort Hotel records were present.

**6.2 Data Quality Assessment**

The dataset was examined for missing values, duplicate records and unusual observations.

Missing values were identified in variables such as company, agent, country and children. The absence of company or agent information may indicate that a booking was not associated with a particular company or travel agent. These missing values therefore require contextual interpretation rather than automatic replacement.

Duplicate records were also examined. The analysis identified exact duplicate rows, which require consideration before any removal because identical records do not necessarily establish that two bookings belong to the same customer.

Additional checks were conducted for unusual stay durations, guest counts and ADR values.

**6.3 Data Transformation and Feature Creation**

Additional variables were created to support the analysis.

These included:

- Arrival date, constructed from arrival year, month and day.
- Total nights, calculated by combining weekday and weekend nights.
- Lead-time groups, used to compare cancellation behaviour across booking intervals.
- Stay categories, based on the relative number of weekday and weekend nights.
- Room-type mismatch indicator, comparing reserved and assigned room categories.
- Previous cancellation history, based on the number of previous cancellations.
- Arrival season and year-month variables for temporal analysis.
- Estimated lodging value, calculated using ADR multiplied by total nights.

These derived variables allowed more detailed comparisons and visualisations.

**6.4 Exploratory Data Analysis**

The analysis was organised into six major areas:

1. Data loading, inspection and preprocessing.
2. Univariate analysis of booking and customer characteristics.
3. Comparative analysis of City Hotel and Resort Hotel.
4. Cancellation analysis.
5. Time-series analysis and seasonality.
6. Revenue management and ADR analysis.

Statistical summaries and visualisations were used to identify distributions, patterns, differences and potential relationships between variables.

### 6.5 Analytical Techniques Used

The following analytical techniques were applied to examine the hotel booking dataset and identify meaningful patterns:

1. **Descriptive Statistics:** Mean, median, mode, standard deviation, variance, minimum, maximum and percentiles were used to understand numerical variables such as lead time, ADR and length of stay.
2. **Frequency and Percentage Analysis:** Booking counts and percentages were calculated for categorical variables such as hotel type, customer type, meal preference, market segment, distribution channel and deposit type.
3. **Grouping and Aggregation:** Pandas grouping and aggregation operations were used to calculate summary statistics and compare booking characteristics across different categories.
4. **Comparative Analysis:** City Hotel and Resort Hotel were compared using booking volume, cancellation rates, ADR, lead time, length of stay and customer characteristics.
5. **Correlation Analysis:** Pearson and Spearman correlation coefficients were calculated to examine the relationship between ADR and lead time, as well as ADR and total nights.
6. **Time-Series and Seasonal Analysis:** Arrival dates were used to analyse monthly booking demand, cancellation rates, ADR, yearly booking patterns and seasonal variations.
7. **Data Quality Assessment:** Missing values, duplicate records, unusual stay durations, guest counts and potentially invalid or unusually high ADR values were examined to assess data reliability.

These techniques helped organise the dataset, summarise important variables, compare booking patterns and derive meaningful insights from the available data.

## 7. Analysis and Findings

### 7.1 Booking Distribution by Hotel Type

The dataset contains more City Hotel bookings than Resort Hotel bookings.

| Hotel Type | Bookings | Percentage |
|---|---|---|
| City Hotel | 79,330 | 66.45% |
| Resort Hotel | 40,060 | 33.55% |
| **Total** | **119,390** | **100%** |

City Hotel accounts for approximately two-thirds of the booking records. Its booking volume is almost twice that of Resort Hotel.

This indicates a greater representation of City Hotel bookings in the dataset.

### 7.2 Booking and Customer Characteristics

The univariate analysis examined booking lead time, ADR, stay duration, guest composition, meal preferences, customer types, market segments, distribution channels, special requests, booking changes and deposit arrangements.

The analysis found that:

- Online TA was the largest market segment, with 56,477 bookings.
- TA/TO was the largest distribution channel, with 97,870 bookings.
- The majority of bookings had no special requests: 70,318 bookings, representing 58.90%.
- No Deposit was the most common deposit category, accounting for 104,641 bookings or 87.65%.
- Transient customers formed the largest customer category in both hotel types.
- Reserved room type A was the most frequently recorded reserved room category, with 85,994 bookings.

These results indicate that online travel agencies and travel-agent/tour-operator channels played an important role in the booking distribution represented by the dataset.

**7.3 Comparative Analysis of City Hotel and Resort Hotel**

The two hotel categories were compared using booking volume, cancellation rates, lead time, ADR, stay duration, guest composition and customer preferences.

- **Booking volume:** City Hotel recorded 79,330 bookings, while Resort Hotel recorded 40,060.
- **Cancellation rate:** City Hotel had a cancellation rate of 41.73%, compared with 27.76% for Resort Hotel. The difference was approximately 13.96 percentage points.
- **Average lead time:** City Hotel bookings had an average lead time of 109.74 days, while Resort Hotel bookings had an average lead time of 92.68 days.
- **ADR:** City Hotel recorded a mean ADR of 105.30 and a median ADR of 99.90. Resort Hotel recorded a mean ADR of 94.95 and a median ADR of 75.00.
- **Length of stay:** The average total stay was 2.98 nights for City Hotel and 4.32 nights for Resort Hotel.
- **Family-oriented bookings:** Bookings containing children or babies represented 6.81% of City Hotel bookings and 9.81% of Resort Hotel bookings, according to the project's family-booking classification.
- **Special requests:** Resort Hotel recorded an average of 0.62 special requests per booking, compared with 0.55 for City Hotel.

Overall, City Hotel had higher booking volume, longer average lead time, higher ADR and a higher cancellation rate. Resort Hotel bookings had longer average stays and a higher proportion of family-oriented bookings.

**7.4 Cancellation Analysis:** Cancellation analysis was a major component of the project.

Out of 119,390 bookings:

| Booking Status | Number of Bookings | Percentage |
|---|---|---|
| Not Cancelled | 75,166 | 62.96% |
| Cancelled | 44,224 | 37.04% |

The overall cancellation rate was **37.04%**.

- **Cancellation by Hotel Type:** City Hotel recorded a cancellation rate of 41.73%, whereas Resort Hotel recorded 27.76%. This indicates a higher cancellation proportion among City Hotel bookings.

- **Cancellation by Lead Time:** Cancellation rates increased across the lead-time groups:

| Lead-Time Group | Cancellation Rate |
|---|---|
| 0–7 days | 9.63% |
| 8–30 days | 27.86% |
| 31–90 days | 37.70% |
| 91–180 days | 44.71% |
| 181–365 days | 55.45% |
| More than 365 days | 67.66% |

Bookings made much earlier were associated with higher cancellation rates. This pattern suggests that reservations with longer booking horizons may involve greater uncertainty.

- **Cancellation by Market Segment:** The highest cancellation rate among the major market segments was observed in Groups, at 61.06%. Online TA recorded 36.72%, while Offline TA/TO recorded 34.32%.

- **Cancellation by Distribution Channel:** TA/TO recorded a cancellation rate of 41.03%, compared with 17.46% for Direct bookings and 22.08% for Corporate bookings.

  **Cancellation by Deposit Type:** The Non Refund category had a particularly high cancellation rate in the dataset. This result should be interpreted carefully because the category may reflect booking arrangements and reservation-status recording practices rather than a straightforward causal relationship.

| Deposit Type | Cancellation Rate |
|---|---|
| No Deposit | 28.38% |
| Non Refund | 99.36% |
| Refundable | 22.22% |

- **Cancellation by Customer Type:** Transient bookings recorded a cancellation rate of 40.75%. Contract bookings recorded 30.96%, Transient-Party bookings 25.43%, and Group bookings 10.23%.

- **Cancellation and Special Requests:** Bookings with zero special requests had a cancellation rate of 47.72%. The cancellation rate was lower among bookings with one or more special requests in the observed data.

- **Repeat Guests and Previous Cancellation History:** Repeat guests had a cancellation rate of 14.49%, compared with 37.79% for non-repeat guests. Bookings associated with a history of previous cancellations showed a cancellation rate of approximately 91.6%, based on the notebook's grouping. This suggests that previous cancellation behaviour may be an important indicator of future booking risk.

- **Reserved and Assigned Room Types:** The analysis compared the reserved room type with the assigned room type. Bookings with matching room types had a cancellation rate of 41.56%, while mismatched bookings had a rate of 5.38%. This difference should not be interpreted as proof that room reassignment prevents cancellations. The timing and operational circumstances behind room assignments are not established by this dataset.

- **Cancellation by Country:** Portugal (PRT) had the largest number of bookings, with 48,590 records, followed by the United Kingdom (GBR) with 12,129 and France (FRA) with 10,415. Among the countries with the largest booking volumes, Portugal recorded a cancellation rate of 56.64%. Country-level cancellation comparisons should account for differences in booking volume.

**7.5 Time-Series Analysis and Seasonality**

The arrival date was constructed using the arrival year, month and day. The resulting date range in the notebook was 1 July 2015 to 31 August 2017.

- **Monthly Booking Demand:** August recorded the highest booking volume, with 13,877 bookings. January recorded the lowest, with 5,929 bookings.

- **Monthly Cancellation Rates:** June recorded the highest monthly cancellation rate at 41.46%, while January recorded the lowest at 30.48%.

- **Monthly ADR:** August recorded the highest average ADR at 140.11, while January recorded the lowest at 70.36. The results indicate that booking demand and ADR varied considerably across arrival months.

- **Weekday and Weekend Stays:** Across the months examined, average weekday nights were generally higher than average weekend nights. July and August showed relatively high average weekday and weekend stay durations.

- **Arrival Day and Week Number:** The 17th day of the month had the highest number of recorded arrivals, with 4,406 bookings. The 31st had the lowest, with 2,208 bookings. Week 33 recorded the highest arrival volume by week number, with 3,653 bookings. Week 53 had the lowest, with 905 bookings. These week-number results combine observations across years and should not be interpreted as a year-specific trend.

- **Year-Month Booking Trends:** Booking volumes were grouped by year and month to examine patterns across 2015, 2016 and 2017. The dataset covers partial years at the beginning and end of the observation period, so comparisons should account for the different periods available in each year.

- **Lead Time and Seasonality:** Average lead time was highest in September at 136.68 days and lowest in January at 45.31 days. By the project's seasonal grouping, average lead time was: Summer had the highest average lead time in this grouping.

| Season | Average Lead Time |
|---|---|
| Winter | 55.52 days |
| Spring | 95.65 days |
| Summer | 128.34 days |
| Autumn | 116.97 days |

**7.6 Revenue Management and ADR Analysis**

Average Daily Rate (ADR) was analysed across hotel types, months, customer categories, lead-time groups, market segments, distribution channels, room types and stay characteristics.

- **ADR by Hotel Type:** City Hotel had a mean ADR of 105.30 and median ADR of 99.90. Resort Hotel had a mean ADR of 94.95 and median ADR of 75.00. This indicates higher typical ADR values for City Hotel in the dataset.

- **ADR by Month, Hotel and Customer Type:** The highest median ADR combination identified was:
  1. Month: August
  2. Hotel: Resort Hotel
  3. Customer Type: Transient
  4. Median ADR: 191.70

  The results suggest that pricing varies across hotel type, month and customer category. Combinations with small booking counts should be interpreted cautiously.

- **ADR and Lead Time:** The Pearson correlation between ADR and lead time was -0.063, while the Spearman correlation was 0.015. Both values indicate a very weak relationship between these variables in the overall dataset.

- **ADR by Market Segment and Distribution Channel:** Median ADR varied across market-segment and distribution-channel combinations. Online TA bookings through TA/TO had a median ADR of 109.80, with 56,153 bookings. Direct bookings through the Direct channel had a median ADR of 105.92, with 12,276 bookings. Some combinations had very small sample sizes, so their median ADR values should not be treated as representative of the wider market.

- **ADR by Reserved Room Type:** Room type H had the highest median ADR at 184.00, based on 601 bookings. Room type A had a median ADR of 89.00 and was the most common category, with 85,994 bookings. The difference in booking volume across room types is important when comparing their pricing patterns.

- **ADR and Length of Stay:** The analysis compared ADR across total-night categories. Average ADR varied across stay durations, with some categories recording higher average values than others. The relationship should not be assumed to be linear, and categories with relatively few bookings require cautious interpretation.

- **ADR by Stay Type:** Weekday-heavy bookings had the highest mean and median ADR among the three categories.

| Stay Type | Mean ADR | Median ADR |
|---|---|---|
| Weekday-Heavy | 103.17 | 95.00 |
| Weekend-Heavy | 99.91 | 93.60 |
| Equal Nights | 95.35 | 89.75 |

- **ADR and Special Requests:** Average ADR increased from 94.96 for bookings with no special requests to 130.57 for bookings with four special requests. The category with five requests had a mean ADR of 127.75, but contained only 40 bookings. This indicates an observed association between special-request count and ADR, although it does not establish a causal relationship.

- **ADR by Cancellation Status:** For City Hotel, mean ADR was 105.75 for non-cancelled bookings and 104.69 for cancelled bookings. For Resort Hotel, mean ADR was 90.79 for non-cancelled bookings and 105.79 for cancelled bookings. The difference between cancelled and non-cancelled booking ADR was more noticeable for Resort Hotel than City Hotel.

- **Estimated Lodging Value:** An estimated booking-level lodging value was calculated using:

  **Estimated Lodging Value = ADR × Total Nights**

- This measure provides a simplified estimate based on the available ADR and stay-duration fields. It is not actual realised revenue because the dataset does not provide a complete record of payments, refunds, fees or revenue received. Cancelled bookings may also have a calculated estimate despite not resulting in a completed stay.

- **ADR Data Quality:** The analysis identified:
  1. 1,959 bookings with zero ADR.
  2. 1 booking with negative ADR.
  3. 3 bookings with ADR greater than 500.

  There were 1,963 bookings with at least one flagged ADR condition, representing approximately 1.64% of the dataset. These observations require contextual review before any decision is made to remove or modify them.

**8. Key Findings**

The major findings of the project can be summarised as follows:

1. City Hotel contributed the larger share of booking records, accounting for 66.45% of the dataset.
2. The overall cancellation rate was 37.04%.
3. City Hotel had a higher cancellation rate than Resort Hotel.
4. Longer booking lead times were associated with higher cancellation rates.
5. Groups had the highest cancellation rate among the major market segments examined.
6. Non Refund bookings showed a very high cancellation rate in the dataset.
7. Repeat guests had a considerably lower cancellation rate than non-repeat guests.
8. August recorded the highest monthly booking volume and highest average ADR.
9. January recorded the lowest monthly booking volume and lowest average ADR.
10. City Hotel had a higher median ADR, while Resort Hotel had a longer average stay.
11. ADR varied across room types, customer categories, market segments and stay patterns.
12. Zero, negative and unusually high ADR values were identified for further data-quality review.

**9. Business Recommendations**

Based on the observed patterns, the following recommendations may be considered by hotel management.

**9.1 Improve Cancellation Management:** City Hotel should review its cancellation patterns and consider reservation-management strategies suited to its higher cancellation rate. Booking lead time, deposit type and customer category may be useful variables for identifying bookings that require closer monitoring.

**9.2 Review Long-Lead-Time Reservations:** The increasing cancellation rates across longer lead-time groups suggest that reservations made far in advance may require appropriate confirmation and communication strategies.

**9.3 Strengthen Customer Retention:** Repeat guests demonstrated a lower cancellation rate than non-repeat guests. Hotels may consider loyalty programmes, personalised communication and improved guest experiences to encourage repeat bookings.

**9.4 Plan for Seasonal Demand:** August showed the highest booking volume and average ADR in the analysis. Hotels may use seasonal booking patterns to support staffing, room availability and pricing decisions.

**9.5 Review Distribution and Market Segments:** Online TA and TA/TO represented important booking sources. Hotels may benefit from evaluating channel-specific cancellation patterns and the commercial performance of different market segments.

**9.6 Use ADR Insights in Pricing Decisions:** ADR differences across hotel types, room categories and months can support more informed pricing strategies. However, pricing decisions should also consider occupancy, costs, competitor prices and actual realised revenue.

**9.7 Improve Data Quality:** Unusual ADR values and missing information should be reviewed systematically. Data cleaning rules should preserve legitimate observations and document any records excluded from future analysis.

**10. Limitations of the Study**

The project has several limitations:

1. The analysis is based on the variables available in the Hotel Booking Demand dataset.
2. Each row represents a booking record and should not automatically be treated as a unique customer.
3. Missing values in variables such as company, agent, country and children may affect selected analyses.
4. The dataset covers a limited historical period, from July 2015 to August 2017.
5. Some categories contain relatively few bookings, making their averages and cancellation rates less reliable.
6. Observational analysis identifies associations and patterns but does not establish causation.
7. ADR-based estimated lodging value is not equivalent to actual realised revenue.
8. The seasonal and lead-time groupings used in the project are analytical classifications and may not represent all operational definitions used by hotels.
9. The dataset does not provide all information required to evaluate hotel profitability, operating costs or complete financial performance.

**11. Conclusion**

This project presented an exploratory data analysis of the Hotel Booking Demand dataset, covering 119,390 booking records from City Hotel and Resort Hotel.

The analysis examined booking characteristics, customer behaviour, hotel-wise differences, cancellation patterns, seasonal demand and ADR. It identified meaningful differences between the two hotel categories. City Hotel recorded a higher booking volume, higher median ADR and higher cancellation rate, while Resort Hotel had a longer average stay and a higher proportion of family-oriented bookings under the project's classification.

Cancellation analysis showed that booking lead time, deposit type, customer type, market segment and previous cancellation history were associated with different cancellation patterns. The time-series analysis highlighted August as the month with the highest booking volume and average ADR in the observed data.

The ADR analysis further demonstrated variation across hotel types, room categories, customer categories and stay patterns. Data-quality checks also identified unusual ADR observations that should be reviewed before further modelling or financial interpretation.

Overall, the project demonstrates how exploratory data analysis can convert raw booking records into useful information about customer behaviour, reservation patterns and hotel pricing. The findings can support more informed decision-making in cancellation management, seasonal planning and revenue management, while recognising the limitations of observational data.

**12. References**


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